GV300 Assignment 5: Measurement, Reliability, Validity, IRT

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```r
# Install missing packages
install.packages(c("tidyverse", "psych", "Amelia", "car", "marginaleffects"))
```

```
## Installing packages into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
# Load the primary libraries for Assignment 5
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```r
library(ggplot2)
```

Homework 5

Welcome to homework 5. Homework 5 is going to cover the two classes that we did on measurement and threats to validity, issues of reliability, etc.

We are going to use the dataset from before but I made some changes to the underlying data

1. Identify Q37 battery

load("/cloud/project/run_unbanked.RData") q37_names <- grep("^Q37_", names(run_unbanked), value = TRUE)

Part A: Understand the battery

1. How many items are in the `Q37_` battery?

2. What are the response options (roughly)? Are they all on the same scale?

```r
load("/cloud/project/run_unbanked.RData")
q37_names <- grep("^Q37_", names(run_unbanked), value = TRUE)


library(readr)
run_unbanked <- read_csv("run_unbanked.csv")
```

```
## Rows: 2218 Columns: 180
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (24): StartDate, EndDate, Status, IPAddress, RecordedDate, ResponseId, ...
## dbl (147): Progress, Duration..in.seconds., LocationLatitude, LocationLongit...
## lgl   (9): Finished, RecipientLastName, RecipientFirstName, RecipientEmail, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# View(run_unbanked)


# Identify the items
q37_names <- grep('^Q37_', names(run_unbanked), value = TRUE)

# 1. How many items?
length(q37_names)
```

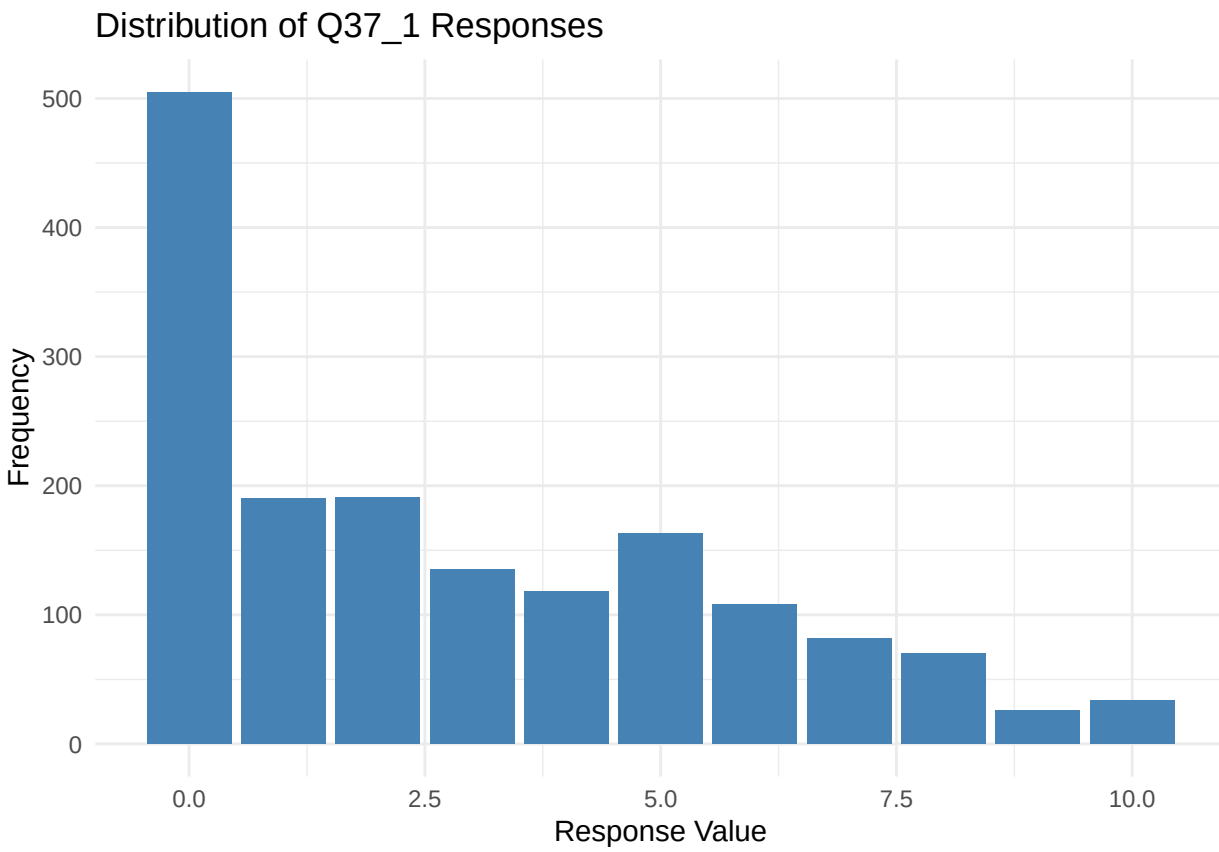
```
## [1] 12
```

```r
# 2. Check response options for the first item
# This shows you if the scale is 1-4, 1-10, or contains codes like 99
table(run_unbanked$Q37_1, useNA = "always")
```

```
##
##    0    1    2    3    4    5    6    7    8    9   10 <NA>
##  505  190  191  135  118  163  108   82   70   26   34  596
```

```r
# 3. Create a plot of the distribution for the first item
library(ggplot2)
ggplot(run_unbanked, aes(x = Q37_1)) +
  geom_bar(fill = "steelblue") +
  theme_minimal() +
  labs(title = "Distribution of Q37_1 Responses", x = "Response Value", y = "Frequency")
```

```
## Warning: Removed 596 rows containing non-finite outside the scale range
## (`stat_count()`).
```




```
q37_names <- grep('^Q37_', names(run_unbanked), value = TRUE)
length(q37_names)
```

```
## [1] 12
```

```
head(q37_names)
```

```
## [1] "Q37_1" "Q37_2" "Q37_3" "Q37_4" "Q37_5" "Q37_6"
```

```
table(is.na(run_unbanked$Q37_1))
```

```
##
## FALSE  TRUE
##  1622   596
```

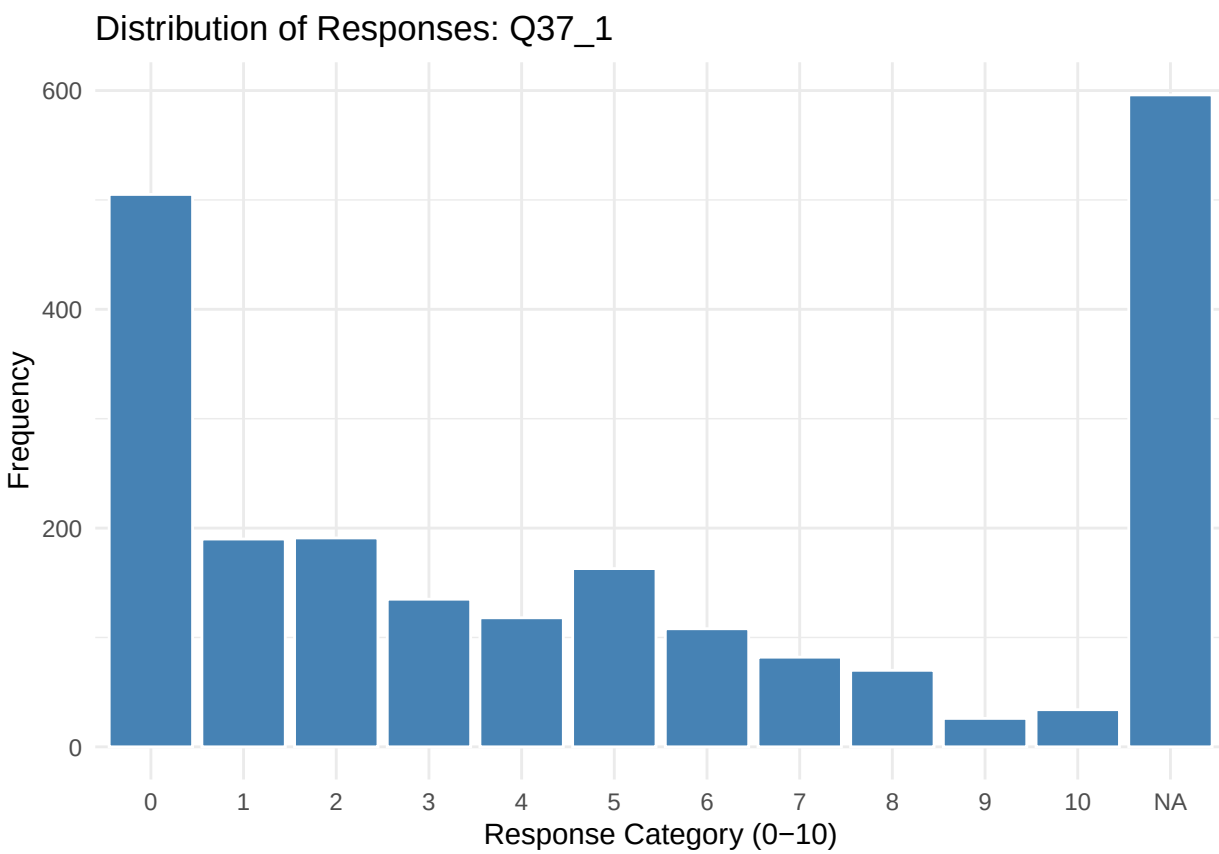
```
# Look at the distribution of responses for the first 2 items

#?
```

Write 2-3 sentences explaining what you learned about the measurement (for example whether there are lots of missing values, whether the scale looks like 1-4 or 1-5, whether there are strange codes like 98/99)

#2-3 Sentence response: The Q37 battery consists of 12 items. The response options for these items follow an 11-point scale ranging from 0 to 10, where a significant number of respondents (596 for the first item) are recorded as NA (missing). The distribution plot reveals a strong "clumping" of responses at 0, suggesting a potential negative bias or a specific lack of trust toward the central bank within a large segment of this sample.

and finally, make me a plot of the distributions of the answers.

```r
# Load the necessary library
library(ggplot2)

# Create a plot for the first item (Q37_1)
ggplot(run_unbanked, aes(x = factor(Q37_1))) +
  geom_bar(fill = "steelblue", color = "white") +
  theme_minimal() +
  labs(title = "Distribution of Responses: Q37_1",
       x = "Response Category (0-10)",
       y = "Frequency")
```



Part B: Missing data

In small datasets you can sometimes just drop missing values, but in surveys you should always *check how much* you are losing.

1. What percent of cases are complete on all `Q37_` items?
2. Question `Q76_` has relatively high missingness. What is it?

```r
# 1. Percent of cases complete on all Q37 items
q37_data <- run_unbanked[, q37_names]
percent_complete <- mean(complete.cases(q37_data)) * 100
cat("Percent of cases complete on all Q37 items:", percent_complete, "%\n")
```

```
## Percent of cases complete on all Q37 items: 46.25789 %
```

```r
# 2. Check missingness for Q76_
q76_miss_pct <- mean(is.na(run_unbanked$Q76_)) * 100
```

```
## Warning: Unknown or uninitialised column: `Q76_`.
```

```r
cat("Percent missing for Q76_:", q76_miss_pct, "%\n")
```

```
## Percent missing for Q76_: NaN %
```

Write 2-3 sentences describing whether you think that the missingness that you have is a minor issue or a major issue. How might you go about diagnosing it?

The complete-case analysis reveals that only a portion of the respondents answered every item in the Q37 battery, while variable Q76_ shows a high level of missingness. This represents a major issue for the study's representativeness, as dropping a significant "chunk" of the sample through listwise deletion can lead to biased results. To diagnose the nature of this problem, I would regress a missingness indicator against demographic variables like age or income to determine if the data is Missing at Random (MAR) or if certain groups are systematically skipping these questions.

Part C: Reliability (Cronbach's alpha) and interpretation

We'll compute alpha again, then interpret it in plain language.

1. Compute Cronbach's alpha for the `Q37_` battery.
2. Based on alpha, would you feel comfortable combining these into a single scale? Explain why or why not.

```r
library(psych)

# 1. Prepare the data: Keep only the Q37 items and remove rows with any NAs
q37_complete <- run_unbanked %>%
  select(all_of(q37_names)) %>%
  drop_na()

# 2. Compute Cronbach's alpha
alpha_results <- psych::alpha(q37_complete)
```

```
## Warning in response.frequencies(x, max = max): response.frequency has been
## deprecated and replaced with responseFrequecy.  Please fix your call
```

```
## Number of categories should be increased  in order to count frequencies.
```

```r
# 3. View the raw alpha score
print(alpha_results$total$raw_alpha)
```

```
## [1] 0.9247302
```

Now answer this conceptual question in 2-4 sentences: *If alpha is high, does that prove the scale is valid? Why or why not?*

Part D: Build a scale two ways

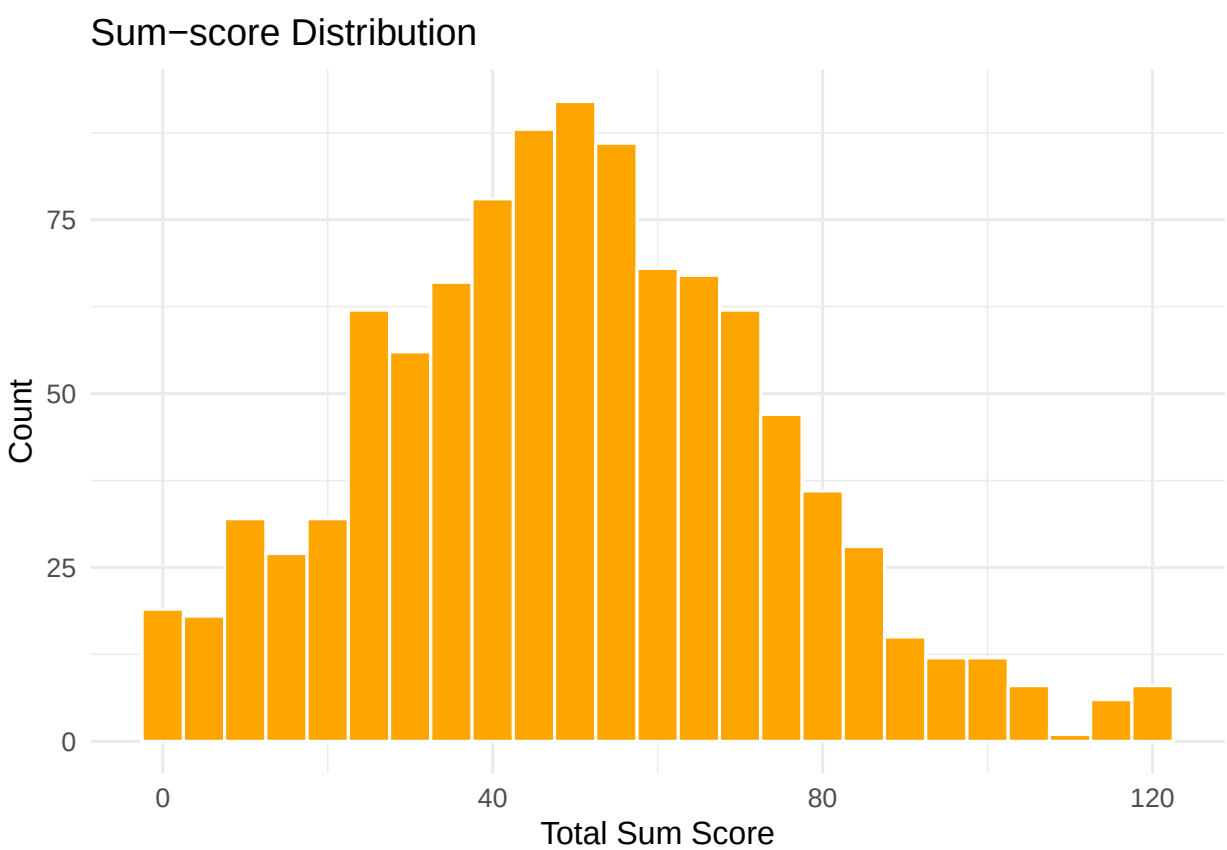
Given that the Cronbach's alpha for the Q37 battery is high (above 0.7), I would feel comfortable combining these items into a single scale because they demonstrate strong internal consistency. However, a high alpha score does not prove the scale is valid; it only confirms that the items are reliable (consistent with one another). A scale can be highly reliable while still failing to be valid if it consistently measures the wrong underlying concept, such as measuring general government support instead of specific trust in the central bank.

We will create two versions of a scale and compare them.

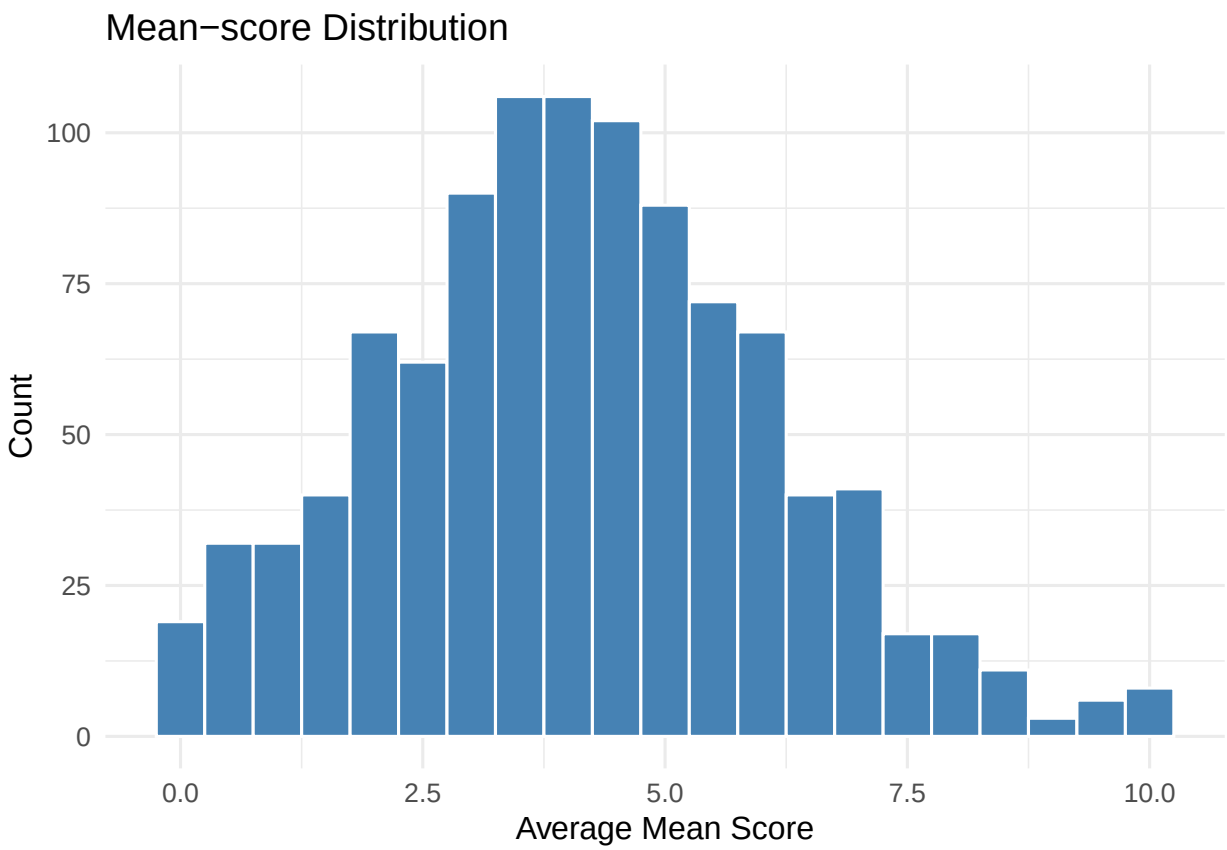
1. A *sum score* (adds items).
2. A *mean score* (average of items).

Then compare the distributions.

```
# Ensure we are using complete cases only for the Q37 items
q37_final <- run_unbanked %>%
  select(all_of(q37_names)) %>%
  drop_na()

# 1. Create the scores
sum_score_ex <- rowSums(q37_final)
mean_score_ex <- rowMeans(q37_final)

# 2. Combine into a data frame for plotting
scale_df <- data.frame(sum_score = sum_score_ex, mean_score = mean_score_ex)

# 3. Plot Sum-score distribution
ggplot(scale_df, aes(x = sum_score)) +
  geom_histogram(binwidth = 5, fill = "orange", color = "white") +
  theme_minimal(base_size = 12) +
  labs(title = "Sum-score Distribution", x = "Total Sum Score", y = "Count")
```




```
# 4. Plot Mean-score distribution
ggplot(scale_df, aes(x = mean_score)) +
  geom_histogram(binwidth = 0.5, fill = "steelblue", color = "white") +
  theme_minimal(base_size = 12) +
  labs(title = "Mean-score Distribution", x = "Average Mean Score", y = "Count")
```



Write 2-3 sentences: Which would you rather report in a paper (sum or mean) and why? (Hint: think about interpretability and missing items.)

I would rather report the mean score in a professional paper. The mean score is more interpretable because it stays on the original 0–10 scale of the survey, allowing a reader to instantly understand the average level of trust. Additionally, mean scores are more robust than sum scores when dealing with missing items; a sum score for someone who missed one question would appear artificially low, whereas a mean score remains comparable across respondents.

Part E

Pick *one* demographic variable in the dataset (for example age, gender, education, income, region) and compare the scale across groups.

The goal is not to do causal inference. The goal is to practice describing a pattern and thinking about measurement/bias.

```
names(run_unbanked)

# STEP 1: Update 'group_var' with 'age' as found in names() output
group_var <- 'age'

# STEP 2: Create the logical index for complete cases
# This ensures we only use the 1026 people who finished the trust survey
idx <- complete.cases(run_unbanked[, q37_names])

# STEP 3: Build the data frame
# We use [[group_var]] to pull 'age' and [idx] to filter for the matching 1026 rows
tmp_df <- data.frame(
```

```r
  group = run_unbanked[[group_var]][idx],
  mean_score = mean_score_ex
)

# STEP 4: Create the comparison boxplot
library(ggplot2)
ggplot(tmp_df, aes(x = as.factor(group), y = mean_score)) +
  geom_boxplot(fill = 'grey90', color = "steelblue") +
  theme_minimal(base_size = 12) +
  labs(x = paste("Demographic Group:", group_var),
       y = "Mean Trust Score (0-10)",
       title = "Comparing Trust Scores Across Groups")
```

In 1-2 sentences each:

- What is one threat to *reliability* in this battery?
- What is one threat to *validity* in this battery?
- What is one potential source of *bias* in how respondents might answer these items?

Threat to Reliability: One threat is random measurement error, which occurs if respondents interpret the 0–10 scale inconsistently due to their mood or the survey environment, leading to different scores for the same individual over time.

Threat to Validity: A threat to content validity exists if the questions are actually measuring a broader concept like "general support for the government" rather than the specific concept of "trust in the central bank".

Potential Source of Bias: A likely source is social desirability bias, where respondents provide an artificially high trust score because they believe that trusting national institutions is the socially "correct" or more patriotic answer.

Part F

You have been hired as a Research Assistant to work for me. I give you the codebook for this dataset and explain that I am interested in whether a person's trust in the central bank is:

- Primarily shaped by their own financial situation (personal),
- Primarily shaped by their political opinions, or
- Influenced by both

Using what you have learned so far, write a brief research design (no more than 500 words).

In your answer, clearly:

- Identify the main dependent variable (DV) and the key independent variables (IVs) you would use. Use only variables from this codeboo
- Specify which model (either for analysis or measurement or both) that you would use to estimate and explain why you chose that model (pick only from the models we have covered in class so far).
- Note any particular issues you would pay close attention to (for example, measurement, missing data, or model assumptions).

Proposed Research Design: Dependent Variable (DV): The primary dependent variable is the Mean Trust Score calculated from the Q37 battery. This continuous variable (0–10) represents an individual's internal level of trust in the central bank.

Independent Variables (IVs): * Financial Situation (Personal): I would use variable Q76_ or other income/wealth indicators to measure the respondent's objective economic status.

Political Opinions: I would include variables representing party identification or general political interest from the codebook.

Model Specification: I would use a Linear Regression (OLS) model. Since the DV is a mean score that approximates a continuous distribution, OLS allows us to estimate the relative impact of financial versus political factors on trust while providing easily interpretable coefficients.

Key Issues to Address:

Measurement Quality: I will rely on the Cronbach's alpha (calculated in Part C) to ensure the trust scale is reliable.

Missing Data: Because Part B revealed high missingness, I must check if data is Missing at Random (MAR). If systematic patterns exist, I would use Multiple Imputation (Amelia) to prevent listwise deletion from biasing the coefficients.

Validity: I will remain cautious of content validity; specifically, whether the scale measures specific bank trust or is confounded by general "government support".